

Qualification Pack



IT Software Solutions for Business

QP Code: SSC/Q0111

Version: 1.0

NSQF Level: 4.5

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SSC/Q0111: IT Software Solutions for Business

Brief Job Description

Individuals at this job are responsible for development of software applications and interfaces as well as enhancements to existing packaged applications or pre-engineered templates. The job also involves providing support to custom applications, debugging, maintenance and documentation. This job requires the individual to work independently and be comfortable in making decisions pertaining to his/her area of work. This job may require the personnel to work as an individual contributor or manage a team of junior contributors.

Personal Attributes

This job requires the individual to have thorough knowledge of various technology trends and processes as well as have updated knowledge about IT initiatives. He/she should be highly motivated and energetic with the ability to self-direct daily activities.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N0102: Work Organization and Management for IT Software Solutions for Business](#)
2. [SSC/N0103: Communication and Interpersonal Skills for IT Software Solutions for Business](#)
3. [SSC/N0104: Problem Solving Innovation, and Creativity](#)
4. [SSC/N0105: Analysis and Design of Software Solutions](#)
5. [SSC/N0106: Development of Software Solutions](#)
6. [SSC/N0107: Testing Software Solutions](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	IT Support Services
Occupation	IT Support Services/Helpdesk
Country	India
NSQF Level	4.5

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Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ NIL
Minimum Educational Qualification & Experience	Completed 1st year of UG (UG Certificate) OR Pursuing 1st year of UG OR Previous relevant Qualification of NSQF Level (4) with 1-2 Years of experience relevant filed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	19 Years
Last Reviewed On	NA
Next Review Date	28/02/2026
NSQC Approval Date	08/02/2024
Version	1.0
Reference code on NQR	QG-4.5-IT-01802-2024-V1-NASSCOM
NQR Version	1.0

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SSC/N0102: Work Organization and Management for IT Software Solutions for Business

Description

Work Organization and Management for IT Software Solutions for Business

Scope

The scope covers the following :

- Work Organization and Management for IT Software Solutions for Business

Elements and Performance Criteria

Work Organization and Management for IT Software Solutions for Business

To be competent, the user/individual on the job must be able to:

- PC1.** Plan each day's production schedule according to available time and take into account time limitations and deadlines
- PC2.** Apply research techniques and skills to keep up-to-date with the latest industry guidelines
- PC3.** Review own performance against the expectations and needs of clients and organizations
- PC4.** Work within the sustainability guidelines expected by the clients and organizations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work Organization and Management for IT Software Solutions for Business</i>	15	15	-	-
PC1. Plan each day's production schedule according to available time and take into account time limitations and deadlines	3	3	-	-
PC2. Apply research techniques and skills to keep up-to-date with the latest industry guidelines	4	3	-	-
PC3. Review own performance against the expectations and needs of clients and organizations	4	4	-	-
PC4. Work within the sustainability guidelines expected by the clients and organizations	4	5	-	-
NOS Total	15	15	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0102
NOS Name	Work Organization and Management for IT Software Solutions for Business
Sector	IT-ITeS
Sub-Sector	IT Support Services
Occupation	IT Support Services/Helpdesk
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

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SSC/N0103: Communication and Interpersonal Skills for IT Software Solutions for Business

Description

Communication and Interpersonal Skills for IT Software Solutions for Business

Scope

The scope covers the following :

- Communication and Interpersonal Skills for IT Software Solutions for Business

Elements and Performance Criteria

Communication and Interpersonal Skills for IT Software Solutions for Business

To be competent, the user/individual on the job must be able to:

- PC1.** • Use literacy skills to:
- Follow documented instructions from supplied guides
 - Interpret workplace instructions and other technical documents
 - Interpret and understand systems specification documents
 - Keep up-to-date with latest industry guidelines
- PC2.** • Use oral communication skills to:
- Discuss and offer suggestions regarding system specifications
 - Keep clients updated regarding systems' progress
 - Negotiate with clients regarding project budgets and timelines
 - Gather and confirm clients' requirements
 - Present proposed and final software solutions
- PC3.** • Use written communications skills to:
- Documents and demonstrates solutions by developing documentation, flowcharts, layouts, diagrams, charts, code comments and clear code.
 - Keep clients updated regarding systems' progress
 - Confirm that created applications meet original specifications and obtain user sign-off for completed systems
- PC4.** • Use team communication skills to:
- Collaborate with others to develop required outcomes
 - Contribute to group problem solving
- PC5.** • Use project management skills to:
- Prioritize and schedule tasks
 - Allocate resources to tasks.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communication and Interpersonal Skills for IT Software Solutions for Business</i>	15	15	-	-
PC1. • Use literacy skills to: • • Follow documented instructions from supplied guides • • Interpret workplace instructions and other technical documents • • Interpret and understand systems specification documents • • Keep up-to-date with latest industry guidelines	2	2	-	-
PC2. • Use oral communication skills to: • • Discuss and offer suggestions regarding system specifications • • Keep clients updated regarding systems' progress • • Negotiate with clients regarding project budgets and timelines • • Gather and confirm clients' requirements • • Present proposed and final software solutions	3	2	-	-
PC3. • Use written communications skills to: • • Documents and demonstrates solutions by developing documentation, flowcharts, layouts, diagrams, charts, code comments and clear code. • • Keep clients updated regarding systems' progress • • Confirm that created applications meet original specifications and obtain user sign-off for completed systems	2	3	-	-
PC4. • Use team communication skills to: • • Collaborate with others to develop required outcomes • • Contribute to group problem solving	3	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC5. • Use project management skills to: • • Prioritize and schedule tasks • • Allocate resources to tasks.	5	5	-	-
NOS Total	15	15	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0103
NOS Name	Communication and Interpersonal Skills for IT Software Solutions for Business
Sector	IT-ITeS
Sub-Sector	IT Support Services
Occupation	IT Support Services/Helpdesk
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

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SSC/N0104: Problem Solving Innovation, and Creativity

Description

Problem Solving Innovation, and Creativity

Scope

The scope covers the following :

- Problem Solving Innovation, and Creativity

Elements and Performance Criteria

Problem Solving Innovation, and Creativity

To be competent, the user/individual on the job must be able to:

- PC1.** • Use analytical skills to:
- • Synthesize complex or diverse information
 - • Determine the functional and non-functional requirements of specifications
- PC2.** • Use investigation and learning skills to:
- • Obtain user requirements (e.g. interviews, questionnaire, document search and analysis, joint application design, and observation)
 - • Research encountered problems independently
- PC3.** • Use problem-solving skills to:
- • Identify and resolve problems in a timely manner
 - • Gather and analyse information skilfully
 - • Develop alternatives for decision making, select the most appropriate alternatives and produce the required solutions
 - • Develop business logic and computational algorithms for specific tasks

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Problem Solving Innovation, and Creativity</i>	20	40	-	-
PC1. • Use analytical skills to: • • Synthesize complex or diverse information • • Determine the functional and non-functional requirements of specifications	5	15	-	-
PC2. • Use investigation and learning skills to: • • Obtain user requirements (e.g. interviews, questionnaire, document search and analysis, joint application design, and observation) • • Research encountered problems independently	5	15	-	-
PC3. • Use problem-solving skills to: • • Identify and resolve problems in a timely manner • • Gather and analyse information skilfully • • Develop alternatives for decision making, select the most appropriate alternatives and produce the required solutions • • Develop business logic and computational algorithms for specific tasks	10	10	-	-
NOS Total	20	40	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0104
NOS Name	Problem Solving Innovation, and Creativity
Sector	IT-ITeS
Sub-Sector	IT Support Services
Occupation	IT Support Services/Helpdesk
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

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SSC/N0105: Analysis and Design of Software Solutions

Description

Analysis and Design of Software Solutions

Scope

The scope covers the following :

- Analysis and Design of Software Solutions

Elements and Performance Criteria

Analysis and Design of Software Solutions

To be competent, the user/individual on the job must be able to:

- PC1.** • Analyse systems using:
- • Use Case modelling and analysis (e.g. Use Case Diagram, Use Case Description, Actor Description, Use Case Package)
 - • Structural modelling and analysis (e.g. Object, Class, Domain Class Diagram)
 - • Dynamic modelling and analysis (e.g. Sequence Diagram, Collaboration Diagram, State Diagram, Activity Diagram)
 - • Data modelling tools and techniques (e.g. Entity Relationship Diagram, Normalization, Data Dictionary)
- PC2.** • Design systems using:
- • Class Diagram, Sequence Diagram, State Diagram, Activity Diagram
 - • Object design and package
 - • Relational or object database design
 - • Human-computer interface design
 - • Security and controls design
 - • Multi-tier application design

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Analysis and Design of Software Solutions</i>	50	100	-	-
PC1. - Analyse systems using: • Use Case modelling and analysis (e.g. Use Case Diagram, Use Case Description, Actor Description, Use Case Package) • Structural modelling and analysis (e.g. Object, Class, Domain Class Diagram) • Dynamic modelling and analysis (e.g. Sequence Diagram, Collaboration Diagram, State Diagram, Activity Diagram) • Data modelling tools and techniques (e.g. Entity Relationship Diagram, Normalization, Data Dictionary)	25	50	-	-
PC2. - Design systems using: • Class Diagram, Sequence Diagram, State Diagram, Activity Diagram • Object design and package • Relational or object database design • Human-computer interface design • Security and controls design • Multi-tier application design	25	50	-	-
NOS Total	50	100	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0105
NOS Name	Analysis and Design of Software Solutions
Sector	IT-ITeS
Sub-Sector	IT Support Services
Occupation	IT Support Services/Helpdesk
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

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SSC/N0106: Development of Software Solutions

Description

Development of Software Solutions

Scope

The scope covers the following :

- Development of Software Solutions

Elements and Performance Criteria

Development of Software Solutions

To be competent, the user/individual on the job must be able to:

- PC1.** Develop software solutions by studying information needs, conferring with users, and studying systems flow, data usage, and work processes
- PC2.** Use database management systems to construct, store and manage the data for the required systems
- PC3.** Use latest software development environments and tools to modify existing codes and write new codes of client-server-based software solutions
- PC4.** Evaluate and integrate appropriate libraries and frameworks into the software solutions
- PC5.** Build multi-tier applications
- PC6.** Construct web enabled or native mobile interfaces for client-server-based systems

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Development of Software Solutions</i>	50	100	-	-
PC1. Develop software solutions by studying information needs, conferring with users, and studying systems flow, data usage, and work processes	10	15	-	-
PC2. Use database management systems to construct, store and manage the data for the required systems	10	15	-	-
PC3. Use latest software development environments and tools to modify existing codes and write new codes of client-server-based software solutions	10	15	-	-
PC4. Evaluate and integrate appropriate libraries and frameworks into the software solutions	10	15	-	-
PC5. Build multi-tier applications	5	20	-	-
PC6. Construct web enabled or native mobile interfaces for client-server-based systems	5	20	-	-
NOS Total	50	100	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0106
NOS Name	Development of Software Solutions
Sector	IT-ITeS
Sub-Sector	IT Support Services
Occupation	IT Support Services/Helpdesk
NSQF Level	4.5
Credits	6
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

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SSC/N0107: Testing Software Solutions

Description

Testing Software Solutions

Scope

The scope covers the following :

- Testing Software Solutions

Elements and Performance Criteria

Testing Software Solutions

To be competent, the user/individual on the job must be able to:

- PC1.** Plan testing activities (e.g. unit testing, volume testing, integration testing, and acceptance testing)
- PC2.** Design test cases with data and check results of test cases
- PC3.** Debug and handle errors
- PC4.** Report on test processes
- PC5.** The implementation of black and white box testing

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Testing Software Solutions</i>	30	50	-	-
PC1. Plan testing activities (e.g. unit testing, volume testing, integration testing, and acceptance testing)	6	10	-	-
PC2. Design test cases with data and check results of test cases	6	10	-	-
PC3. Debug and handle errors	6	10	-	-
PC4. Report on test processes	4	10	-	-
PC5. The implementation of black and white box testing	6	10	-	-
NOS Total	30	50	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0107
NOS Name	Testing Software Solutions
Sector	IT-ITeS
Sub-Sector	IT Support Services
Occupation	IT Support Services/Helpdesk
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- Carry out assessments under realistic work pressures found in the normal industry workplace (or simulated workplace).
- Ensure that the range of materials, equipment, and tools that learners use are current and of the type routinely found in the normal industry workplace (or simulated workplace) environments.
- Assess that all tasks and functions are completed in a way, and to a timescale that is acceptable in the normal industry workplace.
- Assign workplace (or simulated workplace) responsibilities that enable learners to meet the requirements of the NOS.
- Productivity levels must be checked to ensure that it reflects those that are found in the work situation being replicated.
- Provide situations that allow learners to interact with the range of personnel and contractors found in the normal industry workplace (or simulated workplace).

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0102.Work Organization and Management for IT Software Solutions for Business	15	15	0	0	30	5
SSC/N0103.Communication and Interpersonal Skills for IT Software Solutions for Business	15	15	0	0	30	5
SSC/N0104.Problem Solving Innovation, and Creativity	20	40	0	0	60	10
SSC/N0105.Analysis and Design of Software Solutions	50	100	0	0	150	30
SSC/N0106.Development of Software Solutions	50	100	0	0	150	40
SSC/N0107.Testing Software Solutions	30	50	0	0	80	10
Total	180	320	-	-	500	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.